



A new index of perceived inflation: Assumptions, method, and application to Germany

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Abstract

After the introduction of Euro notes and coins in January 2002, throughout the Economic and Monetary Union member countries a substantial discrepancy was evident between inflation as measured by the official consumer price indices (CPI) and that perceived by the general public. The starting point of this paper is the German case. First, the public controversy in Germany after the Euro changeover is sketched. Then conventional approaches to perceived inflation are reviewed including the many studies published by the German Federal Statistical Office as well as the *perceived inflation balance* generated within the EU Consumer Survey. In the main part of the paper, a novel approach to measuring perceived inflation is developed, the *Index of Perceived Inflation* (IPI). First, the hypotheses underlying this index are presented. Then, the IPI is derived. In the fourth section the IPI is applied to the German data. The IPI time series for Germany from 1996 through 2005 shows a particularly high perceived inflation around the introduction of Euro notes and coins. In the fifth section the hypotheses of inflation perception underlying the IPI are critically reviewed. The paper closes with a conclusion on the insights gained through computation of the IPI.

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1. Introduction

After the introduction of the Euro in January 2002, a substantial discrepancy was evident throughout the Economic and Monetary Union member countries between inflation as measured by the official consumer price indices (CPI) and that perceived by the general public. This was also true for Germany: there was a widespread feeling that the introduction of the Euro as legal tender had brought about a significant hike in consumer prices. Amidst the general uproar, Consumer Protection Minister Künast even organised an “Anti-Teuro Summit” (“teuro” is a play on words linking the word “euro” with the word “teuer”, meaning expensive). The German Chancellor also shared the consumers’ concern. In 2002, “teuro” was voted word of the year.

No, the Euro was no *teuro*, retorted the official statistics and leading economics research institutes. It was pointed out that the consumer price index in May 2002 had risen by only 1.1% over the previous twelve months and that it was at its lowest level since November 1999 (1.0%). The European inflation rate, at 2.0%, exhibited the same level in the twelve months to May as had been recorded just before the introduction of Euro notes and coins. One could truly not speak of a price hike.

The Ifo Institute for Economic Research in Munich emphasised that the introduction of the Euro had been a “non-event” in terms of inflation and ascribed the debate over the *teuro* to the “imaginary realm of the psyche and subjective perceptions” (Sinn, 2002). Other influential economic and political opinion leaders dismissed the phenomenon as well. In May 2005, the *Frankfurter Allgemeine Zeitung* (2005), a major German daily, stated that the *teuro* had simply been a mere illusion for many. The oh-so-naïve consumer had just been affected by a “distorted price feeling”. What was not there just could not be.

The gap between perceived and measured inflation is an important phenomenon in various respects (see Del Giovane & Sabbatini, 2005). It can lead to erosion of the Euro’s public acceptance and its institutional framework, signs of which had already become evident in Italy in 2005. Furthermore, this gap may influence inflation expectations and thus lead to distorted prices and wages. It can also lead to questioning the credibility of monetary policy, based as it is upon a price index that is not publicly accepted. Lastly, this gap possibly diminishes the capacity of producers and consumers to assess prices correctly, reducing thereby the price system’s allocative efficiency.

This makes evident that measuring perceived inflation is a useful undertaking. However, until late 2004 no method to directly quantify perceived inflation was known. In September 2004, Brachinger proposed a concept for an index of perceived inflation (“IPI”) in the *Neue Zürcher Zeitung* (Brachinger, 2004). This concept was subsequently stated more precisely (Brachinger, 2005b) and presented at several international congresses (e.g., Brachinger, 2005c). The purpose of this index is a direct measurement of the annual perceived inflation rate that is not based on survey data, but on directly or indirectly available official data on price changes.

In a joint project between the CESTat Center for Research in Economic Statistics at the University of Fribourg in Switzerland and the German Federal Statistical Office (“Destatis”) the IPI has been calculated for Germany. This index made it possible for the first time to examine the gap between official and perceived inflation more thoroughly. This article presents the principal findings of that project.

The following section gives a brief account of the analyses carried out by the Federal Statistical Office and presents the usual EU indicator to measure perceived inflation.

The third section of this paper addresses the perceived inflation issue and examines the problem of measurement. First, it introduces briefly the hypotheses of inflation perception developed by Brachinger. Then, on the basis of these, the index of perceived inflation (“IPI”) is developed. In the fourth section follows a presentation of the basic findings of the project conducted jointly with Destatis. It is shown that the IPI indeed well traces inflation as perceived by the German general public. At the time of the introduction of Euro notes and coins, this index was several times higher than the official Consumer Price Index (CPI). In the fifth section the hypotheses of inflation perception underlying the IPI are discussed. The paper closes with a conclusion on the insights gained through computation of the IPI.

2. Conventional approaches to perceived inflation

2.1. Analyses by the federal statistical office

The German Federal Statistical Office recognised quite early on the gap between CPI and perceived inflation and presented several studies that addressed this matter (see for instance Buchwald, Chlumsky, & Engelhardt, 2002; Chlumsky & Engelhardt, 2002). From mid-2001, a total of 18,000 price series of selected products of daily needs from 35 product groups were examined together with the Deutsche Bundesbank (German Central Bank).

It was indeed possible to identify a series of “special developments” (see Statistisches Bundesamt, 2002a) which, according to the Statistical Office, could have contributed to the consumer’s perceived inflation. The first of these was that on the occasion of the currency changeover the number of price changes had increased markedly. In the opinion of the Statistical Office, this was caused by the many “Euro-induced” price changes, i.e. those due to the migration from an attractive price in DM to an attractive price in Euros. A second “special development” was that, at the moment of introducing Euro notes and coins, significant price hikes were observed in certain services (such as hair dressing and dry cleaning) that were not reversed later on.

In addition, at the beginning of the year 2002 a significant price increase was observed in foodstuffs. In January 2002, foodstuffs were 6.7% more expensive than in the same month a year before. However, the price increases observed for foodstuffs, had basically been attributable to a harsh winter in the southern European fruit- and vegetable-producing areas. In November 2002 the price level of foodstuffs, with a year 2000 = 100 basis, was already 0.9% lower year-on-year, while fruit and vegetables were 2.2% and 5.3% cheaper, respectively.

Conversely, the prices of many industrial products, in particular those related to entertainment electronics, had fallen significantly. Thus, computers indeed cost 23% less in May 2002 than in the same month the year before. The net effect of all these price movements had been negligible. On top of this, the number of price changes throughout the year 2002 had decreased continuously and by December 2002 had again reached its usual long-term level. Lastly, most services had shown below-average price increases in 2002 in comparison with the year before.

All things considered, the Statistical Office estimated the influence of all Euro-induced price changes ascertained from October 2001 to October 2002 upon the annual rate of inflation for the product groups examined to lie between -1.6% and $+2.7\%$ points (see

Statistisches Bundesamt, 2002a). This is a minimal influence, if one considers that the product groups examined account for only a small portion of the *CPI*.

The findings of these efforts always delivered the same message: the introduction of the Euro as legal tender had no significant influence upon the German *CPI*. The annual index, purged of special effects such as tax increases and price hikes of fruit and vegetables caused by weather conditions was already 1.6% in January, which was the trend of the previous months. From mid-year on, the inflation rate levelled off even more, with inflation on a basis of 2000 = 100 being 1.3% in October 2002 and 1.2% in November 2002.

In short: No satisfying explanation could be presented for the enormous gap between *CPI* and perceived inflation in the years from 2002 to 2004. Destatis – just as every other European statistical office – faced this phenomenon more or less helplessly and has stressed repeatedly since then that inflation in the period after the introduction of Euro notes and coins was in no way higher than that prior to its introduction.

In July 2004, Destatis released a “definitive” study on the issue of whether the Euro was indeed a *teuro* (Statistisches Bundesamt, 2004). According to it, in the last two-and-a-half years of the Deutschmark (DM) inflation in Germany had clocked 4.3%, while in the first two-and-a-half years of the Euro it had reached only 3.3%. Thus, it was hoped, the matter could be laid to rest once for all. The *Süddeutsche Zeitung* regarded the “*teuro* feeling” as unquestionably disproved (Einecke, 2004). The prestigious *Neue Zürcher Zeitung* subsequently consigned the Euro-as-*teuro* notion to the realm of myths (Rasonyi, 2004).

In fact, the public uproar regarding the Euro as *teuro* had lessened in the meantime and it could almost appear that consumers have accepted that their high perceived inflation actually was a distorted perception. “*The days of high-inflation perception are over*”, gushed the (*Frankfurter Allgemeine Zeitung*, 2005 (FAZ)) and asserted that “*the gap between perceived and actual inflation*” had disappeared.

2.2. The EU consumer survey balanced perceived inflation (*BPI*) measure

The assertion of the FAZ had been based on a comparison of the development of the German *CPI* with the time series of the *perceived inflation balance* generated within the EU Consumer Survey. The *Joint Harmonised EU Programme of Business and Consumer Surveys* (European Commission, 2003) is a programme set up by the European Commission to “provide essential information for economic surveillance, short-term forecasting and economic research”. This survey collects inter alia data from a Europe-wide sample of 21,000 consumers. Among other questions it asks respondents how, in their opinion, consumer prices have developed over the last 12 months. Have they risen a lot (PP), risen moderately (P), risen slightly (E), stayed about the same (M), fallen (MM), or do you not know (N)?

On the basis of the distribution of the various answers (PP + P + E + M + MM + N = 100%), an *aggregate balance* is calculated. The idea of this *balanced perceived inflation measure* (*BPI*) is to quantify the “discrepancy” between the “positive” and the “negative” side of the answers. This balance is calculated as the weighted average

$$BPI = \left(PP + \frac{1}{2}P \right) - \left(\frac{1}{2}M + MM \right). \quad (1)$$

Obviously, the *BPI* oscillates between a minimum value of −100, when all respondents choose the most negative option and a maximum value of +100, when all respondents

choose the most positive option. The development of the BPI time series for Germany from January 1995 to July 2005 is shown by the dotted line in Fig. 1 (left scale). To provide a comparison between perceived inflation and that officially measured, the annual percentage growth rates of the German *harmonised consumer price index* (HCPI) are shown next to the BPI time series (right scale). (The HCPIs of the member states of the European Union (EU) are versions of the national CPIs obtained by homogenising their most significant methodological aspects to facilitate international comparisons.)

It is evident that the BPI time series parallels the CPI until 2002. Then a “gap” between the two time series opens until it closes down again by 2004. It can be readily seen that the BPI time series peaks markedly around the time of Euro introduction and that for 2005 it shows, in comparison to 2002, a clear decrease. From this purely graphic comparison the FAZ derived its assertion of a “disappearing gap” between perceived and actual inflation. This is, however, in no way a correct interpretation of the Consumer Survey aggregate balance.

If the balance BPI of the response categories is high, it suggests, in fact, that the *proportion* of those in the target population who perceived prices as having increased clearly surpasses that of the respondents who perceived the opposite. The higher this balance, the larger this proportion. In this respect, this indicator shows something about the extent of a higher perceived inflation in a target population. But this indicator says nothing regarding the change in the level of perceived inflation itself, measured in percent. Qualitative surveys only provide a direction of change for a given variable, rather than an exact figure.

Hence it is most problematic when – as the FAZ did – the aggregate perceived inflation balance is placed directly against the consumer price index. Both indicators are measured on very different scales and, depending on how both curves are scaled in a graph, an almost arbitrary fit can be portrayed. That means that the scale of the BPI curve can always be trimmed in such a way as to obtain a picture as “near” as possible to the desired

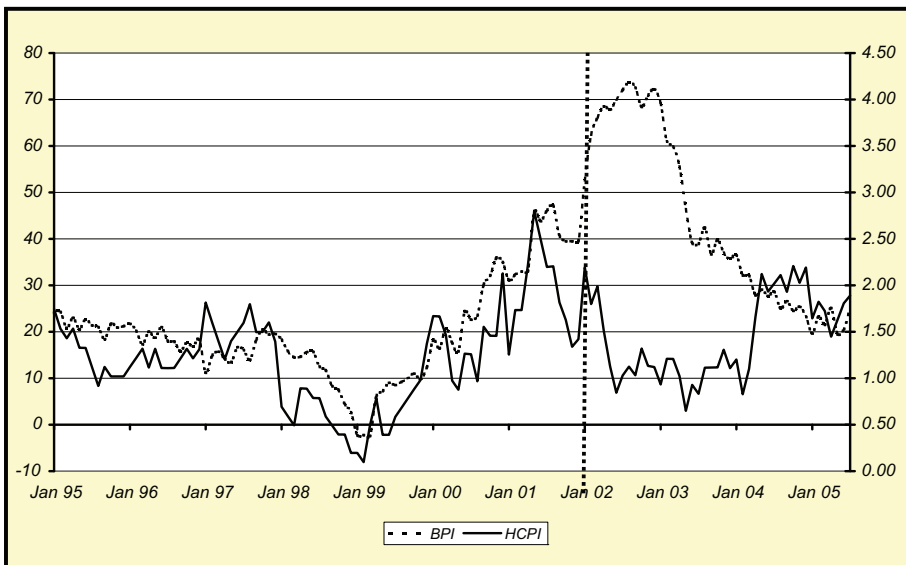


Fig. 1. Annual percentage growth rates of the Harmonised Consumer Price Index (“HCPI”; rescaled right-hand scale) and Balanced Perceived Inflation Measure (“BPI”; left scale) in Germany from 1/95 to 7/05.

picture. This is obviously the case in Fig. 1. In interpreting this figure one may only compare the behaviour of the two curves not their levels. One cannot therefore assert that the gap between the levels of perceived and actual inflation is closing.

A correct analysis of the development of the Consumer Survey perceived inflation balance BPI for Germany shows that, indeed, the difference between $PP + 1/2P$, i.e. the sum of the proportion of those who perceive prices to have risen a lot (PP) and half of the proportion of those who perceive prices to have risen moderately ($1/2P$), and $MM + 1/2M$, i.e. the proportion of those who perceive prices to have stayed about the same (M) or fallen (MM), was diminishing in 2004. On the other side, this balance shows that $PP + 1/2P$ clearly surpasses $MM + 1/2M$ even in the year 2005, B oscillating slightly above 20. In that sense, the *teuro* feeling, on average, remained in fact till 2005 and beyond.

One might retort that this proposition cannot be maintained (see e.g., Hoffmann, Leifer, & Lorenz, 2006 p. 149) because, in the first nine months of 2005, the ranks ‘risen a lot’ and ‘risen moderately’ accounted for just under 48% of the responses on average, while the ranks ‘risen slightly’, ‘stayed about the same’, and ‘fallen’ accounted for 50%. This argument is misleading because it measures the “discrepancy” between the “positive” and the “negative” side of the answers by the simple difference

$$D := (PP + P) - (E + M + MM). \quad (2)$$

This difference, however, does not fulfill the basic properties a balance should have. E.g., it takes on the maximum value of +100 in either cases, when all respondents choose the most positive or the second positive option. It takes on the minimum value of –100, when all respondents choose any of the three remaining options. So this difference is not “balanced”. A “balance” should take on the minimum and the maximum value only for the cases when all respondents choose the most negative option or, respectively, the most positive one. It should be equal to 0 if all the respondents choose the “neutral” option.

The BPI-measure was up to 2005, in fact, the only empirical measure for measuring perceived inflation. However, this measure has some important drawbacks. First, there are no hypotheses about perceived inflation underlying that measure. So this measure has no satisfying explicative power as to explain the gap between officially measured and perceived inflation.

Second, what is needed for a comparison of the magnitude of perceived inflation with the CPI is a price index which measures the rate at which inflation perception changes. For a given number N of commodities with price vectors $p_0 \equiv (p_0(1), \dots, p_0(N))$ and $p_t \equiv (p_t(1), \dots, p_t(N))$ at base period 0 and comparison period t , respectively, a *price index* is a positive function

$$P : \begin{cases} \mathfrak{R}_{++}^{2N} \rightarrow \mathfrak{R}_{++} \\ (p_0, p_t) \mapsto P(p_0, p_t) \end{cases} \quad (3)$$

satisfying a number of certain axioms (see, e.g., Eichhorn (1978)). A particular value $P(p_0, p_t)$ is called a price index number. A price index is a method to compress the information concerning the price changes of a number of commodities into a single number. The axioms assure that a price index provides the economic information one is looking for. As it can take on negative values, obviously by construction, the BPI – measure is not a price index at all.

Various methods have been put forward in the literature for converting qualitative survey data like the BPI – series into quantitative estimates (see e.g., Nardo (2003) and the papers cited there). However, these methods are based on empirically unsubstantiated assumptions such as that perceptions are unbiased, e.g. that in the long run perceived inflation is equal to the CPI. In addition, research such as that conducted by Del Giovane and Sabbatini (2005) shows that the perceived inflation rate inferred varies markedly from one approach to another. Results generated by these methods are therefore also highly non-robust.

What is needed is a perceived inflation measure which is, like the CPI, a price index in the sense of definition (3). For a better understanding of why perceived inflation diverges from that officially measured this index should belong to the same family of indices as the CPI. Different specifications of both indices can then serve as an explication for the divergence of perceived and officially measured inflation.

In the following section, a new index to directly measure perceived inflation will be presented. Unlike the BPI – indicator this measure is based on official price surveys. It is an objective statistical indicator which for comparability reasons belongs to the same class of price indices that most consumer price indices including the German CPI belong to.

3. Theory and measurement of perceived inflation

3.1. Basic hypotheses about inflation perception

The development of an index of perceived inflation requires the specification of some basic hypotheses about inflation perception. In this subsection, three hypotheses first stated by Brachinger (2005c, 2005b, 2005a) will be formulated. Two of these hypotheses are derived from principles that are the core of the famous Prospect Theory developed by Kahneman and Tversky (see Kahneman & Tversky, 1979; Tversky & Kahneman, 1991) as a descriptive model for individual decision-making under risk.

To begin with, it has to be emphasised that something like “true inflation” does not exist. An immediate and unconditional access to inflation is not possible; inflation cannot be measured directly and without any assumptions. The development of any price index requires an interpretative framework of what shall be understood by “inflation”. Any interpretative framework is comprehended as a certain way of seeing inflation and determines not only what is perceived as inflation but also which aspects of inflation should be considered. Interpretative frameworks are prerequisites for the perception of reality and for reflections on reality. As in the context of other economic measurement problems different interpretative frameworks are conceivable.

A first interpretative framework of inflation emanates from the *consumer’s perspective*. Under this perspective, consumption of certain goods during a given period is the subject-matter. Of particular interest are the price changes of the goods consumed. A good is the more important the higher is its proportion of the total costs of consumption. The emphasis is on the costs incurred to an average consumer over the reporting period through the consumption of the goods contained in a market basket. The consumer’s perspective is at the roots of all CPIs worldwide.

A second interpretative framework of inflation arises from the *purchaser’s perspective*. From this perspective the phenomenon of inflation is seen differently. Under this perspective, acts of purchase of certain goods during a given period are the subject-matter. Of particular interest are the price changes of the goods purchased. From the purchaser’s

perspective it appears meaningful, instead of looking at the costs of a market basket of goods, to measure how frequently these goods are purchased. A good is the more important the higher is its proportion of the total number of purchasing acts.

In the following, it is assumed that any attempt to measure perceived inflation has to take on the position of the average purchaser. The starting point of the following hypotheses about perceived inflation, therefore, is the purchaser's perspective on perceived inflation.

In a first basic hypothesis about inflation perception, it is assumed (1) that in a preliminary perception phase the price of each good the buyer is confronted with is, isolated from other goods, encoded as a *gain* or a *loss* with respect to a reference price specific to that good. For instance, a 60-Euro menu does not cost just 60 Euros, but will be perceived as relatively expensive or relatively cheap depending on the reference point. If a patron at a restaurant expects a menu price of 60 Euros (that would be his reference point) and is charged only 45 Euros, he will consider the difference as a "gain". If he, however, had expected a 30-Euro menu and is confronted with a 45 Euro check, he will rate this difference as a considerable "loss". This hypothesis is in accordance with Tversky and Kahneman's *reference dependence assumption* that "carriers of value are gains and losses defined relative to a reference point" (Tversky & Kahneman, 1991 p. 1039).

In a second basic hypothesis about inflation perception it is then assumed (2) that every price change is evaluated according to a value function V , whereby a price increase ("loss") is evaluated higher than a price reduction ("gain"). This means that the buyer reacts much more sensitively to price increases than to price reductions. It is assumed that this asymmetric evaluation is performed individually for every price change the consumer is confronted with. The asymmetric treatment of gains and losses, referred to as *loss aversion*, has been observed by Kahneman and Tversky in a number of experiments of choice under risk and uncertainty (Kahneman & Tversky, 1984; Tversky & Kahneman, 1991).

Finally, according to the purchaser's perspective, it is assumed (3) that purchasers form their inflation judgements by mentally aggregating instances of price changes that they recall. Inflation will be deemed to be higher, the more often the purchaser experiences price increases, thereby making it easier for him to recall from memory examples of noticeable price increases. Accordingly, price reductions of goods purchased seldom or of those goods lacking an explicit purchasing action whose prices are charged once a month such as rent on an apartment, have barely any effect at all on inflation perception. This frequency hypothesis implies that availability, that is, ease of memory retrieval, is a crucial factor in inflation perception. The *availability heuristic* where people base their prediction of the frequency of an event or the proportion within a population on how easily an example can be brought to mind has first been reported by Tversky and Kahneman (1973).

On the basis of these assumptions, to measure perceived inflation, Brachinger (2005c, 2005b) proposed an index of perceived inflation ("IPI"). The IPI starts from actual price data and integrates these data in such a way that the three basic hypotheses on inflation perception are taken account of. This index belongs to a class of price indices which can be understood as price indices of a generalised Laspeyres type. This class of price indices will be introduced in the following section.

3.2. Generalised Laspeyres type price indices

The development of a price index to measure perceived inflation requires an interpretative framework of what shall be understood as "perceived inflation". The understanding

that traditionally underlies the official measuring of inflation in practically all countries of the European Economic and Monetary Union is based upon the idea that, for the purpose of measuring inflation, only the prices of the goods of a market basket are to be considered as variables. The purpose is to capture the *pure price evolution* in an economy. Accordingly, when measuring inflation consumption, i.e. the quantities of the goods consumed are held constant.

This approach appears to be appropriate also for measuring perceived inflation. The individual purchaser is concerned most of all with the extent to which, according to her subjective perception, she is affected by inflation in her daily shopping. The focus of inflation perception lies exclusively with price changes. Accordingly, changes in perception that could be induced by changes in buying frequencies are disregarded.

Last but not least, for the sake of the desirable comparability with the consumer price index, the IPI is based, like practically all consumer price indices the world over, upon the *Laspeyres construction principle* which is characterised by its ease of interpretation. It is based on a market basket of n goods that represents the consumption habits of an average household. Laspeyres type price indices can be generally represented by the so-called mean-value formula

$$P_L^{0,t} = \sum_{i=1}^n \frac{p_t(i)}{p_0(i)} \times \frac{p_0(i)q_0(i)}{\sum_{i=1}^n p_0(i)q_0(i)} \quad (4)$$

where $p_t(i)$ and $p_0(i)$ denote the prices of the i th good for the reporting period t or for the base period 0, respectively, and $q_0(i)$ the quantities consumed by the average household of the good i in the base period. In this formula, the price relatives of all the goods in the market basket are weighted by the corresponding expenditure shares for the base period. A Laspeyres price index shows the factor by which the market basket of goods has become more expensive over the reporting period in comparison with the base period when the price relatives $p_t(i)/p_0(i)$ of each individual good are weighted by the corresponding expenditure shares.

From this formula a class of generalised Laspeyres type price indices is induced by, first, allowing optional reference prices $p_r(i)$ with which actual prices are compared and, furthermore, admitting transformations of the price relatives $p_t(i)/p_r(i)$ as well as discretionary weighting factors. In order for the indices to be of the Laspeyres type, these weighting factors must characterise the conditions in the base period. This leads to the class of *Generalised Laspeyres type Price Indices* given by

$$P_{gL}^{r,t} = \sum_{i=1}^n G_i \left(\frac{p_t(i)}{p_r(i)} \right) \times g_i \quad (5)$$

where G_i is any non-negative real-valued transformation of some price relative of the i th good and the g_i denote discretionary base period weights with which the individual price relatives are weighted. These weights are positive and normalised to one, $g_i \geq 0$, $\sum_{i=1}^n g_i = 1$.

Now, the idea of the index of perceived inflation consists of choosing the reference prices and selecting the transformation function as well as the weights in the generalised Laspeyres index formula in such a way that they take account of the hypotheses about perceived inflation as outlined in Section 3.1. The IPI can then be conceived as a special case

of a price index of the generalised Laspeyres type and can be directly compared with the CPI being obviously also a special case of that same family of price indices.

3.3. Index of perceived inflation (“IPF”)

3.3.1. Reference prices

In the first hypothesis about perceived inflation outlined in Section 3.1 the assumption is made that in a first phase of perception the price of each good with which the average purchaser is confronted is, isolated from other goods, encoded as a gain or a loss with respect to a good-specific reference price. The notion of reference price has become more and more popular in modelling price effects on consumer behaviour in recent years. Various conceptualisations of reference price can be found in the literature. These conceptualisations vary from external conceptualisations like, e.g., an advertised price, to internal conceptualisations like means of previous purchase prices. For an overview of some twenty-six different concepts of reference price see Lowengart (2002).

The purpose of the IPI is a direct measurement of annual perceived inflation rates that is based on directly or indirectly available official data on price changes. This index should make it possible to examine the gap between official and perceived inflation around the introduction of Euro notes and coins and should hence be comparable with the official German CPI. However, the CPI time series refers to prices of the base period. For temporal consistency with the CPI time series, also in the IPI, the prices of the base period are taken as reference prices, $p_r(i) = p_0(i)$.

3.3.2. Value function

In the second hypothesis about perceived inflation outlined in Section 3.1 it is then assumed that gains and losses are evaluated according to a value function V , whereby losses are evaluated higher than gains. One could think of these value functions depending on the type of good and its observed price level. This meant that, within a perceived inflation index of the type $P_{\text{gl}}^{r,t}$, each current price $p_t(i)$ of a given good i should not only be encoded as a gain or a loss respectively through the additive comparison with the corresponding good-specific past price $p_0(i)$ but also be evaluated through a value function V_i depending on the price level.

However, it appears justifiable to assume that value functions are independent of the type of good and its price level. The generalised Laspeyres type price index formula (5), as any price index formula, is a function of price relatives and hence not a function of absolute but of relative price changes as can be seen from the well-known identity

$$\frac{p_t(i)}{p_0(i)} = \frac{p_t(i) - p_0(i)}{p_0(i)} + 1 \quad (i = 1, \dots, n). \quad (6)$$

Therefore, each transformation function G_i can be understood as a function

$$G_i\left(\frac{p_t(i)}{p_0(i)}\right) = G_i\left(\frac{p_t(i) - p_0(i)}{p_0(i)} + 1\right) = G_i^*\left(\frac{p_t(i) - p_0(i)}{p_0(i)}\right) + 1 \quad (7)$$

of relative price changes. That is, the transformation function G_i describing the perception of a price relative p_t^i/p_0^i mainly is determined by a term describing the perception of relative price changes.

Weber (1834) has shown that for many sensory dimensions there is a so-called *just noticeable difference* of the sensory input that is detectable by a human being. This just noticeable difference is an increasing linear function of the base level of the sensory input, i.e., a constant fraction of the existing stimulus. Interpreting these thresholds as measurement units of the subjective sensation lead Fechner (1860) to the well-known *Weber–Fechner Psychophysical Law*: equal relative changes in a stimulus bring about equal absolute changes in perception and absolute changes in perception are a linear function of relative changes in the stimulus.

Assuming that the Weber–Fechner Psychophysical Law also holds for the perception of price changes the term $G_i^*((p_t(i) - p_0(i))/p_0(i))$ can be interpreted as the change in perception induced by a price change $p_t(i) - p_0(i)$. It follows that the perception of a price relative $p_t(i)/p_0(i)$ only depends on the perceived price change and is independent of the price level. Furthermore, perceived absolute price changes are a linear function of relative price changes, i.e., for every i holds

$$G_i^*\left(\frac{p_t(i) - p_0(i)}{p_0(i)}\right) = G^*\left(\frac{p_t(i) - p_0(i)}{p_0(i)}\right) = c\left(\frac{p_t(i) - p_0(i)}{p_0(i)}\right). \quad (8)$$

This implies that for all the transformation functions G_i described in Eq. (7) and for any price relative $p_t(i)/p_0(i)$ finally holds

$$G_i\left(\frac{p_t(i)}{p_0(i)}\right) = G_i\left(\frac{p_t(i) - p_0(i)}{p_0(i)} + 1\right) = c\left(\frac{p_t(i) - p_0(i)}{p_0(i)}\right) + 1. \quad (9)$$

According to Kahneman and Tversky's loss aversion assumption "losses loom larger than gains" (Tversky & Kahneman, 1992 p.298), i.e., "losses and disadvantages have greater impact on preferences than gains and advantages" (Tversky & Kahneman, 1991 p.1039). This implies that the value function is steeper in the negative than in the positive domain. According to that assumption price increases are valued higher than price decreases, and hence the constant c should be higher for price increases than for price decreases. Assuming that this constant is already normalised to 1 for price decreases the transformation function for price relatives following from the above hypotheses is given by

$$G\left(\frac{p_t(i)}{p_0(i)}\right) = \begin{cases} \frac{p_t(i)}{p_0(i)} & \text{for } p_t(i) \leq p_0(i) \\ c\left(\frac{p_t(i) - p_0(i)}{p_0(i)}\right) + 1 & \text{for } p_t(i) > p_0(i). \end{cases} \quad (10)$$

Note that because of

$$c\left(\frac{p_t(i) - p_0(i)}{p_0(i)}\right) = \frac{c(p_t(i) - p_0(i))}{p_0(i)}$$

it follows from the Weber–Fechner Psychophysical Law that the transformation function (10) is based on a value function

$$V(p_t(i)) = \begin{cases} p_t(i) - p_0(i) & \text{for } p_t(i) \leq p_0(i) \\ c(p_t(i) - p_0(i)) & \text{for } p_t(i) > p_0(i) \end{cases} \quad (11)$$

which is independent of the price level $p_0(i)$. In this value function it is assumed that the ratio between perceived price increases and perceived price decreases (the "loss aversion

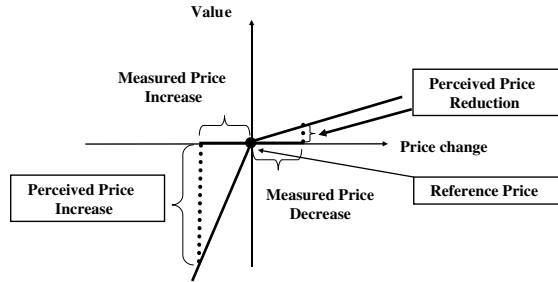


Fig. 2. Typical IPI value function with loss aversion coefficient $c > 1$.

coefficient”) equals c . The loss aversion assumption implies $c > 1$. A typical IPI value function with a loss aversion coefficient $c > 1$ is shown in Fig. 2.

The loss aversion coefficient c can be estimated from perceived price change data. It follows from Eqs. (8) and (11) that this coefficient is, for any fixed (positive) change rate r , given by the ratio between the perception of a price increase at that rate and (the absolute value of) the perception of a price decrease at that same rate. A reasonable estimator for the true perception of a price increase at rate r is then the average over perceived price increases that have been observed for factual price increases at that rate. The average over perceived price decreases that have been observed for factual price decreases at the rate r is a reasonable estimator for the true perception of price decreases of that extent. An estimator of the loss aversion coefficient results from dividing the former average by the latter (see Jungermann, Brachinger, Belting, Grinberg, & Zacharias, 2007 for details).

3.3.3. Weighting scheme

In order to fully specify a price index of the generalised Laspeyres type index formula (5), suitable weights g_i must be selected. The third hypothesis about perceived inflation outlined above assumes that the average purchaser assesses inflation to be higher the easier it is for him to retrieve from memory examples of noticeable price increases. It is expected that the ease with which examples of price increases are recalled depends on the frequency with which the corresponding goods are purchased.

This implies that the frequency of experience of price increases should have a significant influence on perceived inflation: the more often a good or service is bought the more influential it should be on perceived inflation. Therefore, it seems appropriate to take, in the index formula (5), as weight g_i the relative purchasing frequency of good i with which this good is bought by the average purchaser. In order for the index to be of the Laspeyres type, these purchasing frequencies must characterise the conditions during the base period.

3.3.4. The IPI formula

Now, the index of perceived inflation $IPI^{0,t}$ results from taking, in the generalised Laspeyres type price index formula (5), for every price relative $p_t(i)/p_0(i)$ the transformation function G derived above and, as weight g_i , the relative frequency f_i^0 with which the good i is purchased in the base period. The index of perceived inflation (“IPI”) is then defined by

$$\text{IPI}^{0,t} = \sum_{i: p_t(i) > p_0(i)} \left[c \left(\frac{p_t(i) - p_0(i)}{p_0(i)} \right) + 1 \right] f_i^0 + \sum_{i: p_t(i) \leq p_0(i)} \frac{p_t(i)}{p_0(i)} f_i^0 \quad (12)$$

The IPI is an inflation indicator that looks at inflation from the purchaser's perspective. It quantifies the degree to which, in its own subjective perception, the individual purchaser is affected by inflation in its daily shopping.

As the German CPI, which essentially is constructed according to formula (4), the IPI (12) belongs to the class (5) of generalised Laspeyres type price indices. There are two main differences between these indices: First, in the CPI the price relatives of all market basket goods and services are not transformed at all, i.e., the transformation functions are all the identity, $G_i \equiv G \equiv \mathbf{1}$, whereas in the IPI only the price relatives which are lower or equal to unity (decreasing or constant prices) go untransformed. The price relatives which are greater than unity (price increases) are transformed according to (9). Each price relative is, according to identity (6), split off into the growth rate part and unity, then the growth rate part is higher valued by the loss aversion coefficient c and to the higher valued growth rate is eventually re-added unity to end up with a figure which can be interpreted as a growth factor, i.e., a price relative. In other words, price increases are evaluated higher than price decreases through boosting the growth rates by the factor c . The second difference between CPI and IPI is that, as weighting scheme, expenditure weights of an average household are employed in the CPI whereas in the IPI buying frequencies are used. Both indices are of the Laspeyres type because they apply base period weights.

4. Perceived inflation in Germany

4.1. Data base

In a joint project with Destatis, the index perceived inflation (12) has been calculated to measure perceived inflation in Germany around the introduction of Euro notes and coins. In order to ensure comparability with the German consumer price index (CPI), it was considered appropriate for this calculation to choose the same basket of goods and services that underlies the CPI. The CPI market basket with basis 2000 comprises 740 so-called *index positions* representing various categories of goods and services that consumers buy. These index positions are the lowest subcategory of the German CPI for which expenditure weights are available.

The prices used were the prices employed by the official statistics for calculating the CPI. These prices are generally weighted averages of average prices polled in the federal states. In order to calculate the IPI, the price relatives of all goods had to be evaluated by means of the transformation function G defined by Eq. (10). That is, prices that had decreased went directly into the IPI, prices that had increased were transformed according to Eq. 9.

The evaluation of price increases with the transformation function G requires knowing coefficient c , which captures the degree of loss aversion of the average purchaser. This coefficient cannot be observed empirically just like that. We were aware of no published research on the value of this coefficient in the context of price change perception.

The decision-making theory literature contains some indications of how high to set the loss aversion of decision-makers in general. Tversky and Kahneman allude to a loss aversion coefficient “slightly greater than two” resulting from an experiment on individuals’

willingness to pay (Tversky & Kahneman, 1991 p. 1053). They stress that from a series of experimental findings on individual decision-making under risk a loss aversion coefficient of about 2 can be derived. In a lottery experiment with real payments a loss aversion coefficient of 2.5 was found. In a more recent article, Kahnemann points out once again that “estimates of the coefficient of loss aversion ... converge on a value of about 2” (Kahnemann, 2003 p. 164). Hardie, Johnson, and Fader (1993) attempted to measure the loss aversion coefficient within the framework of an experimental study dealing with brand selection. They arrived at the conclusion that this coefficient, depending on the variable observed, lies in an interval between 1.5 and 2.5.

On the basis of these indications, alternative calculations were performed for the computation of the IPI using loss aversion coefficients of 1.5, 2 and 2.5. It is assumed that the IPI(2.0) represents a reasonable estimation of the “true” index of perceived inflation. The band between the IPI(2.5) and IPI(1.5) time series is interpreted in the sense of a “confidence band”, with the “true” IPI lying with high probability within this band.

The main problem with ascertaining the IPI for Germany resided in determining the purchasing frequencies for the 740 index positions. Given that the German CPI takes 2000 as the base year, in order to safeguard both the Laspeyres principle and the comparability between the IPI and the CPI, the purchasing frequencies should be those for the 2000 base period.

The purchasing frequencies turn up as a by-product of the data collection carried out periodically on a random sample of households to determine the expenditure weights for the consumer price index. In Germany these frequencies could be obtained from the Income and Consumption Survey (EVS). They could be taken from the so-called Sampling Notebooks given to the sampled households to help them keep their records but which do not form part of the actual data collection documentation. However, the labour intensive nature of analysing these notebooks made it impractical to use them.

Instead, approximate relative purchasing frequencies were determined from the actual expenditure shares $w_0(i)$ employed with the CPI. It can be seen from the CPI mean-value form

$$P_L^{0,t} = \sum_{i=1}^n \frac{P_t(i)}{P_0(i)} \times \frac{P_0(i)q_0(i)}{\sum_{j=1}^n P_0(j)q_0(j)} = \sum_{i=1}^n \frac{P_t(i)}{P_0(i)} \times w_0(i)$$

that the quantity $q_0(i)$ of good i consumed by the index household during the base period can be determined from the expenditure share $w_0(i)$ of that good, by multiplying the share with the total expenditure of the index household and dividing by the average base period price of that good, i.e.

$$q_0(i) = w_0(i) \times \frac{\sum_{j=1}^n P_0(j)q_0(j)}{P_0(i)}.$$

When interpreting the quantity $q_0(i)$, it must be borne in mind that, underlying these quantities, is precisely the quantity unit that is being priced. This implies that in certain cases the quantities $q_0(i)$ represent very good approximations of the purchasing frequencies. In other cases these quantities overestimate or underestimate the actual purchasing frequencies. In some cases, secondary statistical material had to be used.

For details regarding determination of purchasing frequencies refer to [Bechtold, Elbel, and Hannappel \(2005\)](#).

4.2. Comparison of CPI and IPI

Given the fact that in the IPI price increases are weighted higher than in the CPI, the IPI can be expected to take on values that lie always above those of the CPI. In fact, however, the IPI values depend most of all also on the relative buying frequencies of the individual index positions. A comparison of the expenditure shares with the relative buying frequencies shows that they are hardly correlated. The Bravais–Pearson correlation coefficient is barely 0.0971.

Both the distributions of the expenditure and the frequency weights show the same mean-value, on account of their normalisation to unity. However, the distribution of the relative buying frequencies is less steep on the left than the distribution of the expenditure shares and shows a variance only half as large as that of the expenditure shares distribution. Furthermore, the distribution of the relative frequencies is less strongly peaked, over a shorter range.

The scatter plot of both weight distributions in [Fig. 3](#) shows that nearly all weights move within the 10 tenth-of-one-percent range, with only a few of them standing out from the mass. The four goods with the highest purchasing frequency are “daily newspapers, retail”, “cigarettes”, “beer in liquor store”, and “bread”. These goods are set apart significantly from the rest of the goods on account of their purchasing frequencies. It stands out that three of these goods have only a very low expenditure weight. The good “cigarettes” is the only good that shows a significantly high weight in both weightings. The four goods with the highest expenditure weights are “apartment up to three rooms”, “apartments

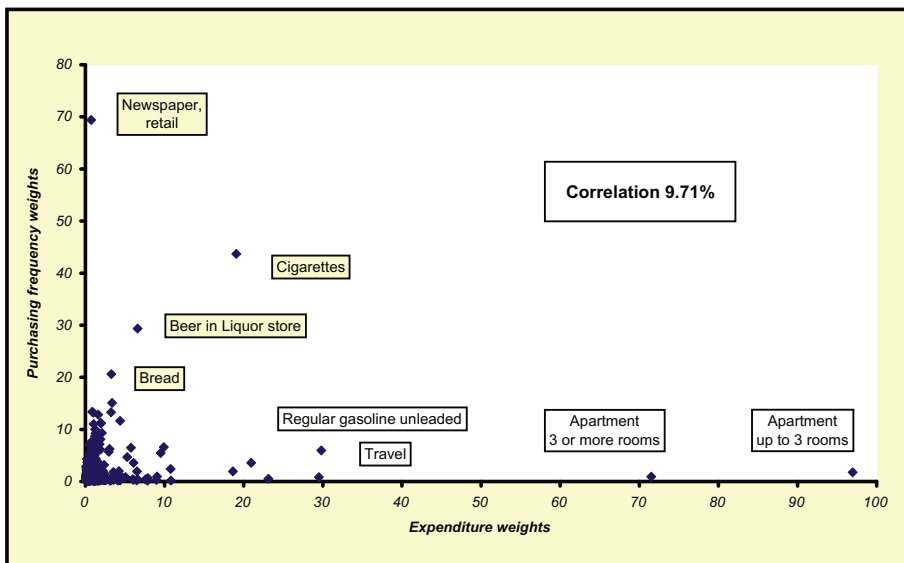


Fig. 3. Scatter plot of expenditure and purchasing frequency weights.

with three or more rooms”, “regular gasoline, unleaded” and “travel”. It is generally noticed that these goods show a very low purchasing frequency.

Considering these quite large differences between the expenditure shares of the CPI and the frequency weights of the IPI it is not surprising that the CPI and the IPI move quite independently of each other. Fig. 4 shows the evolution of the annual percentage growth rates of the CPI and of the IPI with a loss aversion coefficient $c = 2$ since 1996.

Three periods are evident immediately from that chart: a first period prior to the introduction of Euro notes and coins, which lasts until the beginning of the year 2001, a Euro-introduction period from January 2001 until June 2002, and a post-Euro-introduction period that starts in July 2002. In the pre-Euro-introduction period, two sub-periods can be made out: a first one lasting until May 1999 in which perceived inflation lies above the CPI, and a second one that stretches from May 1999 through December 2000. In the second sub-period perceived inflation lies below the CPI. This is all the more remarkable considering that in IPI(2.0) price increases are entered with a double weight compared to CPI!

This figure shows clearly that particularly high perceived inflation can be diagnosed around the point of the introduction of Euro notes and coins. At year-end 2000 CPI and IPI rates still coincided. Afterwards the IPI rates shot up, reaching in June/July 2001 a maximum value of 8.8%, and remained above 7% throughout the year. The CPI rate, in contrast, moved around the 2% mark. In January 2002, at the time of adoption of the Euro notes and coins, the IPI rate jumped to 10.6%. The CPI rate rose as well then, but it increased from 1.6 in December 2001 to a publicly derided 2.1%. In June 2002 the CPI rate settled down again at 1.0% and remained for the rest of the year around one percent. The IPI rate also fell as the year wore on and in July it lay at the level of the CPI rate. The “gap” between perceived and official inflation had closed again.

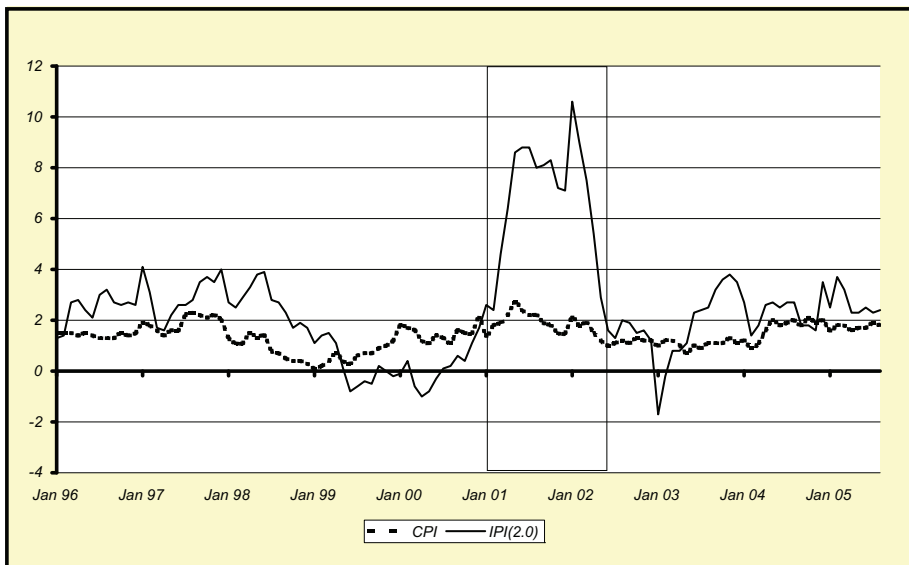


Fig. 4. Comparison of the IPI(2.0) and CPI evolutions.

During the period from January 2001 to June 2002, the average perceived inflation rate, at 6.6%, hovered well above the average rate of the CPI, with 1.8%. The average perceived inflation was then 3.7 times higher than the official inflation rate.

Starting in July 2002, the IPI moved closer to the CPI level. Both peaks in January and November 2003 could be explained by baseperiod effects. The relationship between perceived inflation and CPI “normalised” itself: the average perceived inflation up to August 2005, with 2.1%, lies above the average CPI rate of 1.4%. The former, then, is 1.5 times higher than the official inflation rate. During the period before the introduction of Euro notes and coins, from January 1996, this ratio was 1.3.

What does this show? At the time of the introduction of Euro notes and coins the perceived inflation, as measured according to the IPI standard, was in fact dramatically higher than the CPI. This period of the “Euro-induced” special inflation does not start, however, with the introduction of Euro notes and coins, but during the previous year. In 2001, Destatis had already observed a whopping 5.1% increase in food prices. The Institute for Applied Consumer Research in Cologne² observed the evolution of prices for a vast sample of 700 everyday commercial goods between June 2001 and January 2002. A careful analysis of these price data shows that for 80% of the goods whose price had increased between June 2001 and January 2002, the prices had already been set in 2001. The same holds, albeit to a smaller extent, for the service sector and retail banking. In each of these two sectors, around 50% of the prices raised till January 2002 had been set during 2001.

The retail sector had evidently increased prices markedly before introduction of Euro notes and coins. Only a small proportion of the especially relevant prices was increased further in January 2002. The additional price increase in January 2002 (base 1995 = 100) was induced mostly by the significant price hikes of tobacco goods (+7.4%) and groceries (+6.7%). The increase in the tobacco tax, however, meant that a decision of the legislator was the decisive factor for the price increase of tobacco products, while for foodstuffs the weather-induced fall in harvest was responsible, and not the currency change.

4.3. *Extent of the “Euro-induced” perceived inflation*

Let us assume that the period before the introduction of Euro notes and coins together with the period after its introduction constitute a period during which there was a “normal” relationship between perceived and official inflation. That this assumption is reasonable is shown by the above analyses. Furthermore, let us assume that both the IPI rates and the CPI rates during this period can be considered as outcomes of corresponding stochastic processes. Thus, the average of the IPI rates and the average of the CPI rates can be considered as estimations of the corresponding inflation levels, and the difference between them as an estimation for the “normal” long-term difference between perceived and officially measured inflation.

Correspondingly, let us assume that in the period from January 2001 through June 2002 a special relationship prevailed between perceived inflation and CPI that was induced by the introduction of Euro notes and coins. The above analyses show that this assumption is

² The author thanks Mr. Rainer Wezel of IfaV for making the base data available.

also reasonable. Then, the average of the IPI rates and the average of the CPI rates during this period can be again considered as estimations of the corresponding inflation levels, while the difference between both averages serves as an estimation of the difference between both inflation levels at the time of the introduction of Euro notes and coins. An estimation of the “Euro-induced” special perceived inflation can be obtained by correcting this difference by deducting the “normal” difference between IPI and CPI.

The average IPI rate over the sub-periods prior to and after the introduction of Euro notes and coins was 1.9%, while that for the CPI rates was 1.3%, which yields an estimated “normal” difference between perceived inflation and CPI of 0.6%. The difference between the averages of the IPI rates and the CPI rates over the Euro-introduction period was 4.8%. Subtracting the “normal” difference from this rate yields an estimation of the Euro-induced special perceived inflation of 4.2%.

This estimation depends naturally on the degree of loss aversion assumed for the IPI. The 4.2% value is based on a loss aversion of $c = 2$. A “confidence interval” for this value is obtained by performing the same computation for IPI’s using loss aversion coefficients of $c = 1.5$ and $c = 2.5$. The extent of the “Euro-induced” special inflation as perceived by the consumer is 2.6% for $c = 1.5$, and 5.7% for $c = 2.5$.

Fig. 5 shows the time series of the CPI together with the time series of the IPIs for all three loss aversion coefficients $c = 1.5$, $c = 2.0$ and $c = 2.5$. It can be seen that the index series evolve differently depending on the degree of loss aversion. It is clear that the evolution paths followed by the CPI and IPI in the period before introduction of Euro notes and coins lie quite close to each other. During the Euro introduction period both paths suddenly diverge markedly. During the post-Euro-introduction period, they appear to evolve parallel to each other again.

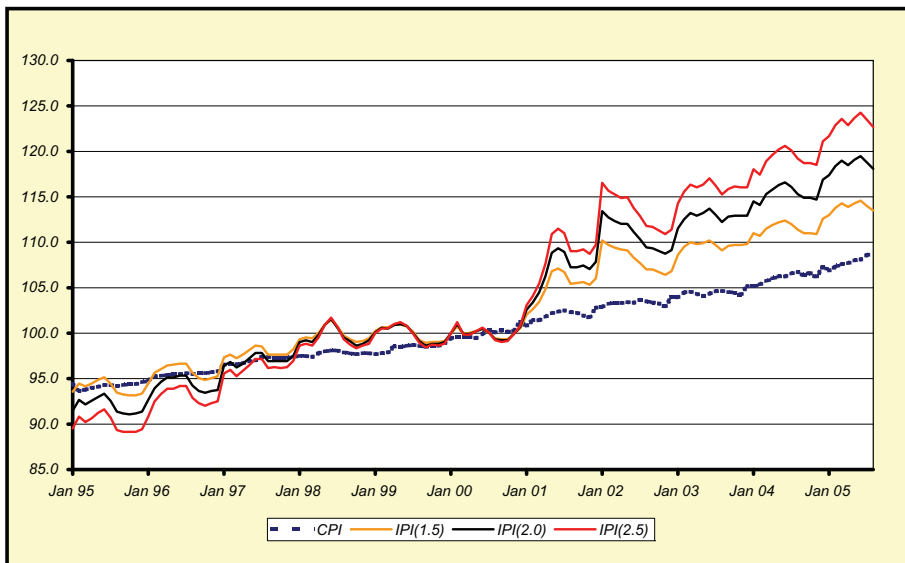


Fig. 5. Comparison of the evolution of CPI and IPI-indices from 1995 to 2005.

An analysis of the CPI and IPI levels during the post-Euro-introduction period makes it possible to estimate the extent to which the general price level, according to public perception induced by the Euro introduction, was at the time above the official price level.

To that, the evolution of the CPI and IPI(2.0) indices will be considered, starting from the beginning of the period defined as “after Euro introduction”, i.e. from July 2002 to the end of the curve in August 2005. The average value of the CPI over this period is 105.6, while the average value of the IPI(2.0) is 114.3, i.e. around 8.2% higher (see Fig. 6). It is assumed then, as was the case above, that the “normal” difference between CPI and IPI levels can be estimated reasonably by the difference of the average CPI and IPI levels during the period before the introduction of Euro notes and coins. These average levels, however, are identical for the IPI(2.0), so that the difference between both average levels in the period after introduction of Euro notes and coins does not need to be corrected. This means that after the introduction of Euro notes and coins, according to public perception, the price level in fact lies around 8.2% higher than shown by the CPI.

For assessing the margin of error induced by the uncertainty regarding the loss aversion coefficient, this figure, again, can be embedded in a “confidence interval”. During the period before introduction of Euro notes and coins the average level of the IPI(2.5) was 0.7 percentage points below the average level of the CPI. Correcting by this value the difference between the average CPI level and the average IPI(2.5) level during the post-Euro-introduction period yields a difference of 12.9% points. Correspondingly, a corrected difference of 4.3% points results for the IPI(1.5). According to public inflation perception, the price level through Euro introduction is only 4.1% higher for the IPI(1.5) but 12.2% higher for the IPI(2.5) than shown by the CPI.

Careful examination of Fig. 6 shows that the evolutions of both price levels do not run parallel to each other. The least-squares fitted straight lines in Fig. 5 indicate that until August 2005 the IPI rises almost twice as fast on average as the CPI.

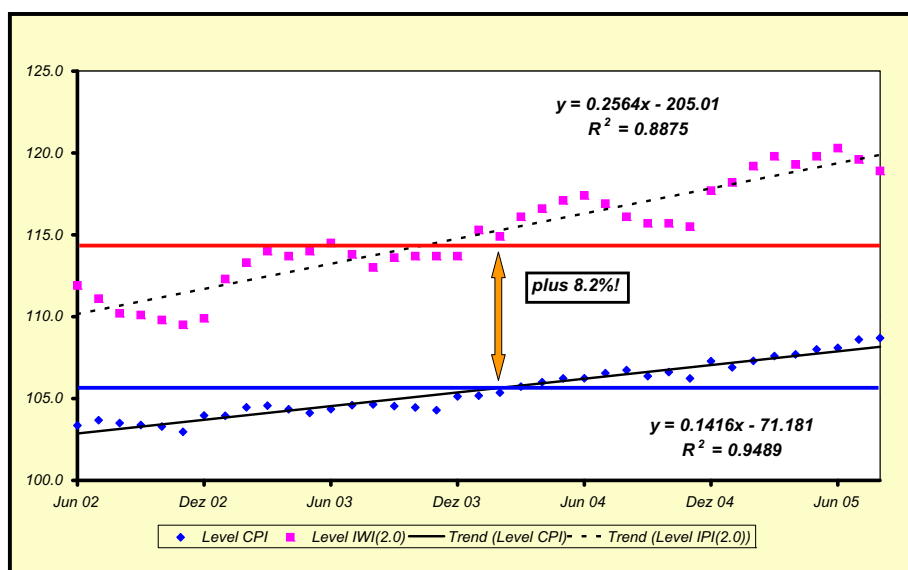


Fig. 6. Comparison of the evolution of CPI and IPI($c = 2$) after June 2002.

5. Discussion of the IPI assumptions

5.1. Reference prices

In Section 3.1 was pointed out that the construction of the IPI is based on the idea that each price the buyer is confronted with is, relative to a reference price, encoded as a gain or a loss. As the IPI is intended to be an index which is comparable with the official CPI, for temporal and structural consistency, the prices of the base period are taken as reference prices relative to which actual prices are compared.

Then the IPI-rates are, as the CPI rates, calculated as annual growth rates of the IPI^{0,t} time series between the actual month t and the same month in the previous year $t - 1$. This rate is finally interpreted as the “true” annual IPI-rate IPI^{t-1,t}. Thereby, it is implicitly assumed that the IPI (approximately) fulfils the *circularity axiom*, i.e., it is assumed that

$$IPI^{0,t} = IPI^{0,t-1} \times IPI^{t-1,t}. \quad (13)$$

Apart from the fact that the IPI, just as the CPI, does not fulfil this axiom, this means that the idea of the annual IPI-rates is that they record the annual growth rate of the inflation as perceived by the average purchaser using the prices of the previous month as reference prices.

However, there is unambiguous evidence that still in 2006 a significant percentage of the EU citizens, when purchasing, did most often count mentally in the old national currency. When it concerns exceptional purchases such as the purchase of a car or a house this percentage was around 40% (cf. European Commission, 2006; Emnid, 2006; Institut für angewandte Verbraucherforschung e.V., 2005; Gesellschaft für Konsumforschung, 2004). This suggests that the familiarity with the old currency has led a considerable portion of consumers to rely for price comparisons on certain base prices that are in some way influenced by prices in the old currency. This *base price rigidity* is an effect which is not typical for general inflation perception, but it is surely an effect that must be regarded characteristic for inflation perception after the adoption of a new currency.

This effect is not modelled by the IPI as defined in Eq. (12) as this index, as reference prices, refers to the prices of the preceding year. Therefore, the IPI index certainly underestimates the degree of inflation perceived by those who, years after the introduction of Euro notes and coins, still calculate in the old currency. For modelling the perception of these people some other prices have to serve as reference prices in the IPI. These reference prices have to be some function of prices in the old currency and Euro prices, e.g., some kind of moving weighted average of such prices. An approach to quantify inflation perception which takes into account the purchasers’ base price rigidity after the Euro changeover is developed in Brachinger (2006).

5.2. Loss aversion

As mentioned at the beginning of Section 3.1, hypotheses (1) and (2), namely that each price the buyer is confronted with is encoded as a gain or a loss and that a price increase (“loss”) is evaluated higher than a price reduction (“gain”) are derived from Prospect Theory developed by Kahneman und Tversky as a descriptive model for individual

decision-making. Some argue (see, e.g., Hoffmann et al., 2006 p. 149) that the loss aversion phenomenon was a phenomenon restricted to the decision-making context. There is however ample evidence that loss aversion is a phenomenon observable in a wide variety of economic domains. These range from labour economics, where downward-sloping of labour supply can be observed (Camerer, Babcock, Loewenstein, & Thaler, 2004), to consumer research, where asymmetric price elasticities have been found (Hardie et al., 1993) (see Camerer, 2004 for an overview).

Anyway it is an interesting question if loss aversion can also be observed in the context of inflation perception and, if yes, if the loss aversion coefficient, in fact, amounts to a value around 2. Up to now, there has been no empirical tests on how price changes are perceived by the average purchaser when judging inflation. In a first experimental study Jungermann et al. (2007) investigated whether the hypotheses formulated in Section (3.1) could be demonstrated in individual judgments of price changes. Participants of the study were asked to estimate price changes of specific products and services. They were given the actual price of a product or service twelve months ago and then asked to estimate the present price, but with three different methods: With method 1, participants were asked to estimate the present price of the product. Different findings show that this method is best suited for measuring perceived inflation. Only this method generated data patterns which were consistent and could be well interpreted.

To estimate the loss aversion coefficient by means of the estimator outlined at the end of Subsection (3.3.2) the perceived price change data produced by all three methods were used separately. To this end, for a certain price change rate, the average over the perceived price increases that have been observed for factual price increases at that rate were divided by the average over the perceived price decreases that have been observed for factual price decreases at that rate. Of course, the estimated loss aversion coefficients depend on the method used to elicit perceived price changes. For method 1 an estimated loss aversion coefficient of 2.02 resulted. The results for the other methods were $c = 2.23$ and 3.07.

These empirical results clearly indicate that, in fact, price increases are evaluated higher than price reductions and, hence, loss aversion is also a phenomenon observable in the context of inflation perception. Furthermore, these figures point up that the choices made for the loss aversion parameter in the IPI calculations for Germany are adequate.

5.3. Psychophysical law

A further assumption the IPI is based on is that the Weber–Fechner Psychophysical Law (Fechner, 1860) also holds for the perception of price changes, i.e., that the perception of price relatives is independent of the price level and a linear function of relative price changes. There is a large body of evidence that this law holds for various specific sensations like, e.g., weight or light (Goldstein, 1996). There is also evidence that it holds for price changes. Tversky and Kahneman (1981), e.g., have noted that a discount of \$5 has a greater impact when the price is low than when the price is high. From the Weber–Fechner Law it follows that the perception of prices is logarithmic which corresponds to the assumption of decreasing marginal returns widely employed in economics. So with the assumption that the Weber–Fechner Law also holds for the perception of price

changes the IPI is in respectable society. In any case, as it is relative price changes that are evaluated, linearity should bring about a sufficiently good approximation.

5.4. *Purchasing frequencies*

The final feature of the IPI is that price changes of the market basket goods and services are weighted by the corresponding relative purchasing frequencies. This frequency hypothesis is based on the idea that purchasers form their inflation judgements by mentally aggregating instances of price changes that they recall. Thereby it is suggested that the ease with which exemplars of increased prices are retrieved from memory depends on personal experience of the buying acts of the average purchaser.

There are many anecdotal examples showing that the connection between recall memory and exposure, generally, is weaker than one might think and sometimes most counter-intuitive. This is also an experimentally validated fact. However, it is also well-known that recall is strengthened by two primary forces: other incidents in which one is forced to recall some information, and the time one spends actively processing a piece of information (e.g. visualising it, spelling it forwards and backwards, etc) (for details see, e.g., [Baddeley, 1997](#)). These two forces both come into play in the context of inflation perception. In the case of a frequently bought good the price change of that good is recalled more often than in the case of a good which is seldom purchased. The time spent actively processing a certain price change, obviously, is longer for frequently bought goods. So the frequency hypothesis seems sensible.

Up to now, there have been no empirical tests of how experience in personal purchases influences perceived inflation. In the above mentioned experimental study ([Jungermann et al., 2007](#)) the authors also tested whether purchase frequency has a significant effect on inflation perception. Their participants were asked to provide on a three-point ordinal scale (“often”, “occasionally”, “never”) their personal judgment of the frequency with which they buy each good presented. [Jungermann et al. \(2007\)](#) calculated an aggregate perceived inflation value for each purchase frequency category by averaging over all participants for each category. They found a significant effect of (subjective) purchase frequency on perceived inflation: perceived inflation is higher for often purchased goods than for occasionally or never purchased goods.

In a another recent experimental study [Huber \(2007, forthcoming\)](#) investigated the hypothesis that availability of price increases determines perceived inflation and that availability depends on experience in personal purchases. The subjects were presented two environments that displayed approximately the same change in total expenditures, and the total expenditures for two groups of products presented in both conditions (cheap with high presentation frequency, and expensive with low presentation frequency) were approximately the same. Reference prices were unambiguously fixed and learned by the subjects, providing a solid base for inflation judgment. The results of [Huber \(2007, forthcoming\)](#) strongly support the frequency hypothesis: whereas in the condition with low frequency of increased prices the inflation judgment was accurate, in the high frequency condition inflation was overestimated by the factor three.

The studies of [Jungermann et al. \(2007\)](#) and [Huber \(2007, forthcoming\)](#) clearly support the frequency hypothesis integrated in the IPI. For further evidence supporting the involvement of availability in inflation perception see Section 5.1 of [Ranyard, Missier, Bonini, Duxbury, & Summers \(2008\)](#).

6. Conclusion

This study was triggered by a publication in the Summer of 2004 in which Destatis (Statistisches Bundesamt, 2004) wanted once and for all to emphasise again all its previous studies that had shown that the Euro was no *teuro*. How could it be that even experienced consumers still in 2004 commented this publication with disbelief?

Section 2 of this article pointed out that no one to date had provided a really satisfying explanation for the immense gap observed between the German consumer price index and perceived inflation in the years after the introduction of Euro notes and coins. The findings presented in this article show that the newly developed index of perceived inflation (IPI) provides a psychologically, economically and statistically well-founded instrument that gives an explanation for this phenomenon.

This index has a psychological foundation because its defining features such as encoding of price changes relative to reference prices, asymmetrical assessment of price changes, and the weighting of price changes by purchasing frequency are derived from well-established facts that are at the core of psychology of decision-making. The IPI shows clearly which factors fuel the perception of inflation. First experimental results like those by Jungermann et al. (2007) outlined in Section 5 suggest that these features, in fact, also play an important role in the perception of inflation.

The index of perceived inflation has an economic foundation because it disregards “moon prices” of goods enthusiastically referred to by certain consumers and news media which, on account of low demand are not purchased at all. Only those prices are considered that, on account of demand for the corresponding goods, have economic relevance. Inflation-distorting subjective price conversions, resulting from a rejective attitude towards the Euro by some consumers, are likewise disregarded.

Finally, the index of perceived inflation has a sound statistical foundation because it is built on a generalisation of a classic formula, the well-known Laspeyres formula. The IPI belongs to the same family of indices as the standard CPI. This assures comparability of IPI and CPI. Reasonably the prices used are the same average prices as used for calculating the CPI. Conversion errors like the one of multiplying Euro prices by 2 in Germany are excluded.

The purpose of the index of perceived inflation is to quantify the subjective, i.e. individual, perception of price increases by an average household. This perception might, with reference to a certain rationality standard, well be considered to be distorted. The index itself however is an *objective* measure as it avoids any form of prejudice, individual bias, or wishful thinking. The rates the index provides are independent from the individual who calculates it. The IPI is, like the CPI, an “incorruptible statistic” (Sinn, 2002).

The analyses presented in this article show that the introduction of Euro notes and coins was not a “non-event”. In the year 2001 something structurally fundamental happened regarding the change in the prices of consumer goods: above-average price increases were applied precisely to those goods characterised by an above-average purchasing frequency. The event thus features a sudden positive correlation between price change and frequency of purchase, something that had not happened previously. As price increases are, compared to price decreases, overvalued a high inflation picture developed.

Why were, however, in particular the prices of goods purchased frequently increased above average prior to the introduction of the Euro? The reasons can here only be speculated on. Destatis has collected indications (see Statistisches Bundesamt, 2002b) that

especially in the retail sector prices were increased significantly in anticipation of the introduction of Euro notes and coins in order to, afterwards, “generously” correct the prices downwards to an “attractive” price in Euros. The evolution of the IPI (see Fig. 4) points however to another explanation: during 1999 and 2000 an extraordinarily low inflation was perceived. During that time there was apparently a significant price war in the retail business that benefited the consumers. Directly afterwards the resulting too low price level was quickly corrected upwards.

Which conclusion should be drawn from the insights gained through computation of the IPI? When a gap appears between the perception of an economic phenomenon and its official measurement, it may not be construed as meaning that one of the sides is wrong or “irrational”. It can well be that, as the case of inflation measurement shows, the official agency measures something different from that what attracts the public interest.

The CPI looks at inflation from the *consumer's perspective*. Under this perspective, consumption of certain goods is of interest. A second interpretative framework of inflation, however, arises from the *purchaser's perspective*. Under this perspective, acts of purchase of certain goods are relevant. Of particular interest are the price changes of the goods frequently purchased. The IPI measures inflation from the purchaser's perspective. So it should not be surprising that both indices, under certain conditions, give a different picture of the same phenomenon.

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